

SCRATCH Pac-Man 1D



Play one-dimensional Pac-Man in Scratch.

1) Download images and create sprites from:

<https://codeclub67.github.io/images/pacman.png>

<https://codeclub67.github.io/images/ghost.png>

<https://codeclub67.github.io/images/pellet.png>

2) Double the size of Pac-Man and the Ghost by setting their **size** to 200%.

3) Set the **direction** of Pac-Man and the ghost to mirror left and right, rather than rotate.

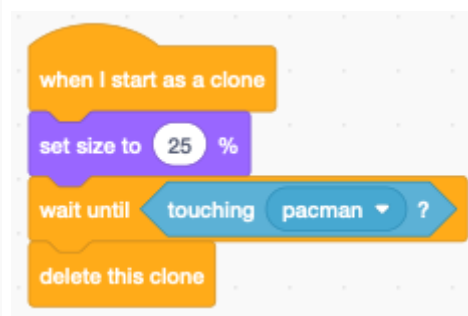
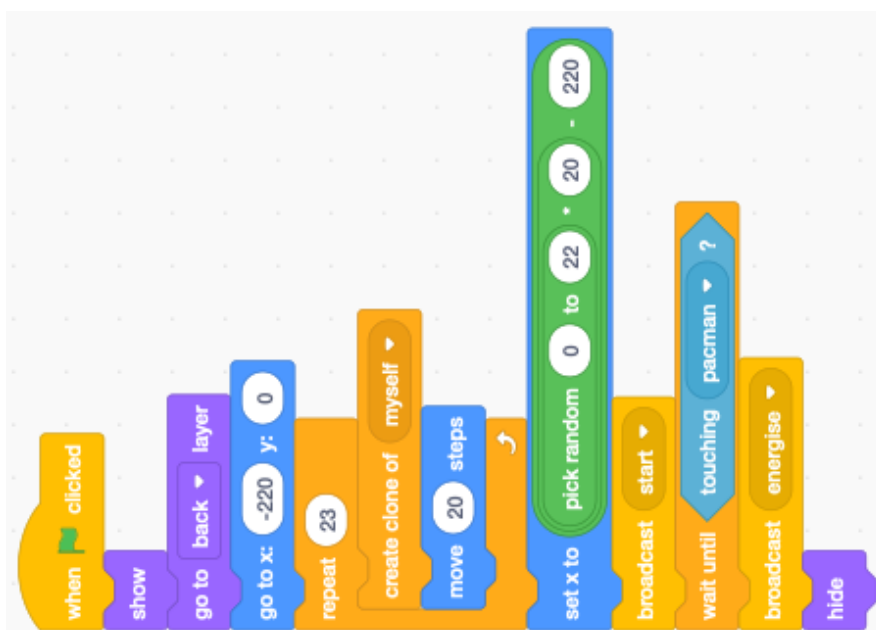


4) Add a dark background (e.g. Stars).

Pac-Man 1D has just a single row of 23 pellets to eat. These small (25% size) pellets are clones.

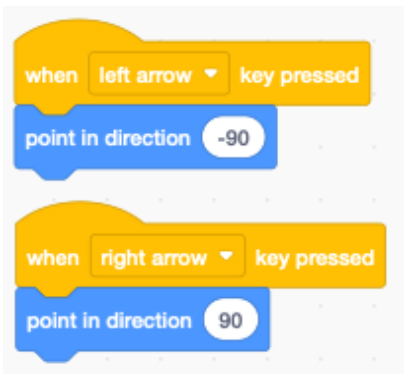
5) Add the code (below) to the **pellet**. The original pellet is

the energiser which is placed **randomly**.



Cycle through Pac-Man's first four costumes to animate it.

6) Add the code below to **Pac-Man**. The **set x** block creates a wrap-around (1D) world for Pac-Man. If it goes off one edge it reappears at the other edge. The grouping of the blocks is $((x \text{ position} + 230) \bmod 460) - 230$.



7) Guide Pac-Man with **left/right keys** (see code left) and point it in the corresponding **direction**.

8) When Pac-Man loses and it's "**game over**", play

out the rest of

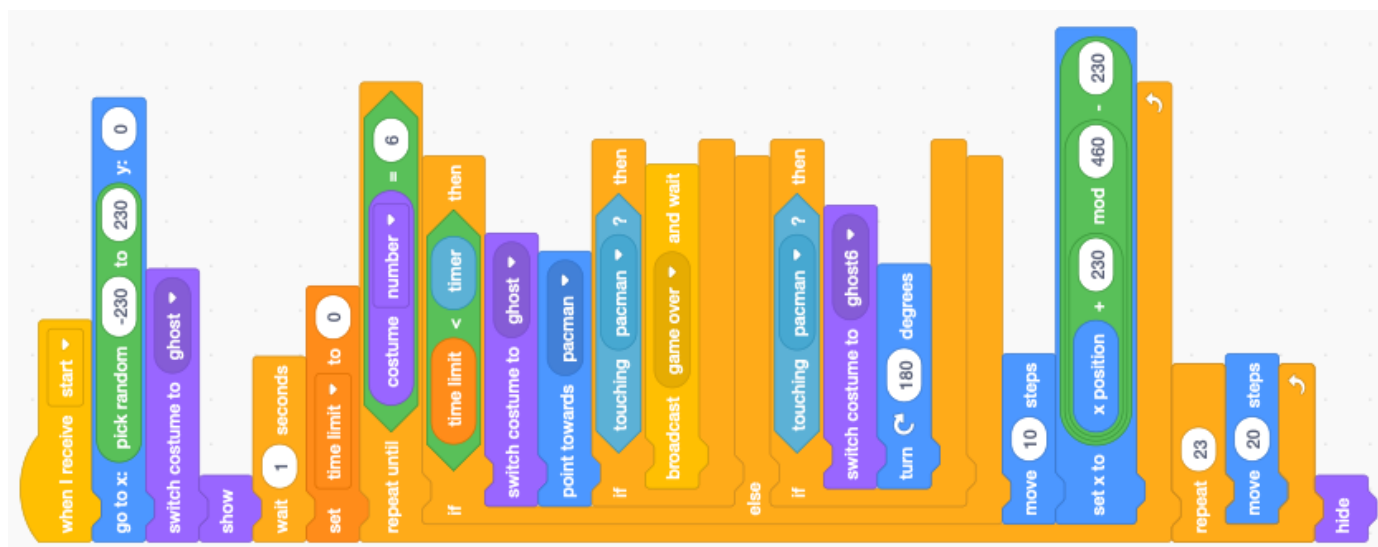
its costumes, up to costume 20, before moving Pac-Man off-screen (see code right).



When Pac-Man is energised it has two seconds to catch the ghost before it goes back to chasing Pac-Man. The energiser has a time limit of 2 seconds into the future using the **timer**. When the **timer** is bigger (later) than the **time limit** the normal ghost costume is displayed, and it heads towards Pac-Man. If it touches Pac-Man it's "game over."

However, if the time limit hasn't expired and the ghost touches Pac-Man, then the ghost 'dies' (costume 6 "eyes").

9) Create a new variable **time limit** and add the code below to the **ghost**. The ghost has a random starting position and it loops until the ghost 'dies'. The small loop at the end animates the 'death' of the ghost.



The set x block does the same wrap-around as in Pac-Man and the grouping of the blocks is the same:

$$(((x \text{ position} + 230) \bmod 460) - 230).$$

The ghost reacts to the energiser by changing to the “frightened” costume and taking a 180° turn away from Pac-Man.

10) Add the code (right) to the **ghost**.

Try duplicating the ghost and selecting a different character (Inky, Blinky, Pinky or Clyde), how hard is it to play?

Save your code with a good name. **File > Save now**

